

WEEK 7 - HEMATOLOGY • SAL HISTOPATHOLOGY LAB • UPDATED WITH
RELEASED REVIEW SLIDES

Lymphoid Tissue Histopathology

Schoenlein review-slide matched • 14-slide released deck • normal lymphoid
architecture

Released SAL images

No dedicated Word companion

Diffuse vs nodular tissue

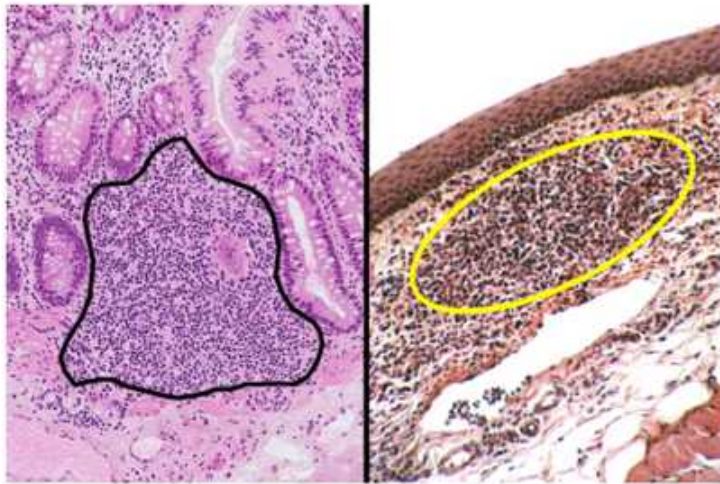
Node / thymus / tonsil / spleen / Peyer's

2-page factoid

SOURCE PRIORITY: the released 14-slide Lymphoid Histology review deck is now the primary source. The prior pre-SAL guide's speculative reactive-lymphadenopathy pattern list has been removed as instructor content because this released deck contains no pathologic lymphadenopathy examples.

Slides 2-4: Diffuse Tissue and Lymphatic Nodules

Diffuse Lymphatic Tissue



Can be solitary or part of larger lymphoid tissue (tonsil, Peyer's patch)

Made up of scattered **lymphocytes** (dark staining nuclei with scant cytoplasm)

Released SAL review slide 2. The review slide says diffuse lymphatic tissue can be solitary or part of larger lymphoid tissue such as tonsil or Peyer's patch, and consists of scattered lymphocytes with dark nuclei and scant cytoplasm.

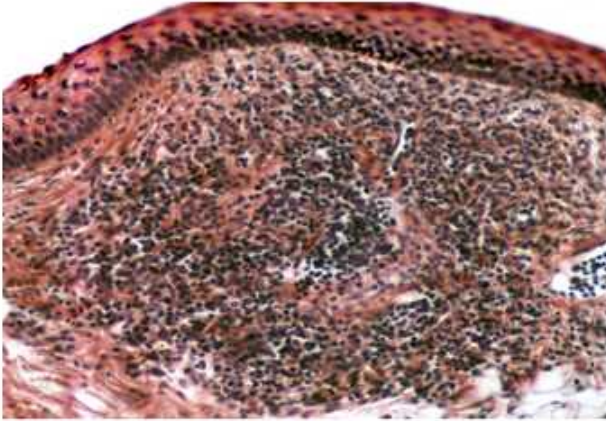
DIFFUSE LYMPHATIC TISSUE

Scattered lymphocytes; dark nuclei + scant cytoplasm. It can stand alone or contribute to a larger lymphoid structure.

Lymphatic Nodules

More organized cells than diffuse tissue

Primary nodule



Lymphocytes (dark staining cells) surrounded/supported by **reticular connective tissue**

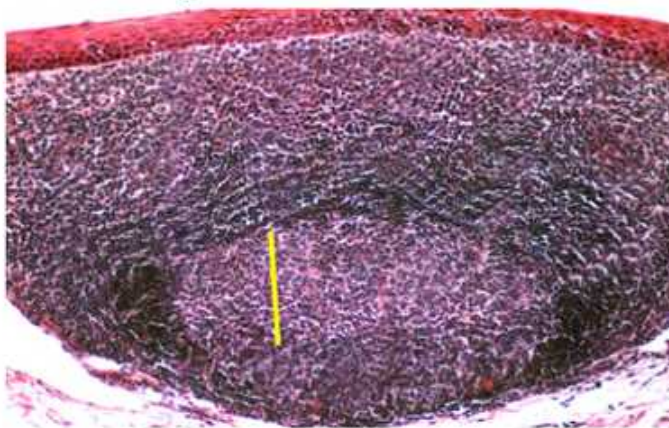
Reticular connective tissue is a loose connective tissue characterized by a delicate, branching network of reticular fibers (Type III collagen) that form a supportive framework, or stroma, for soft organs like the liver, spleen, lymph nodes, and bone marrow, housing other cells and supporting blood vessels and channels. Produced by specialized fibroblasts called reticular cells, this mesh-like structure provides support, filters blood, and allows immune cells to move freely

Released SAL review slide 3. The slide describes a primary nodule as more organized than diffuse tissue. Dark lymphocytes are supported by reticular connective tissue; the slide defines this stroma as a delicate Type III collagen network produced by reticular cells.

Primary nodule: organized dark lymphocytes + reticular connective-tissue support; **no pale germinal center is identified on this slide.**

Lymphatic Nodules

Secondary nodule



Similar to primary, but contains a pale **germinal center** (maturing lymphocytes)

Released SAL review slide 4. The secondary nodule is presented as similar to a primary nodule but with a pale germinal center containing maturing lymphocytes.

PRIMARY VS SECONDARY

Secondary = pale germinal center. That is the slide's fastest discriminator.

Slide 5: Peyer's Patch

Peyer's Patch



Lymphoid tissue in the mucosa of the ileum (distal small intestine)

Made up of lymphoid nodules and diffuse lymphoid tissue

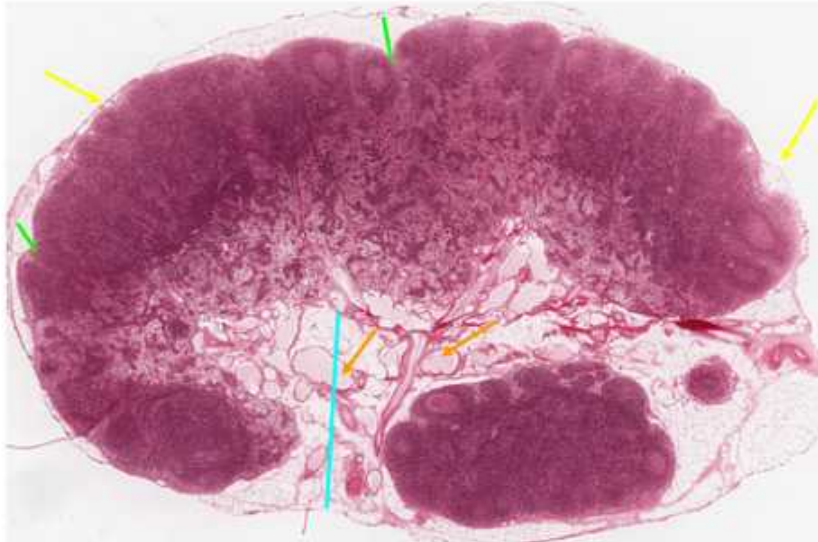
Released SAL review slide 5. The review slide places Peyer's patches in the mucosa of the ileum, the distal small intestine, and describes them as lymphoid nodules plus diffuse lymphoid tissue.

PEYER'S PATCH RELEASED-SLIDE FINGERPRINT

Ileum mucosa + lymphoid nodules + diffuse lymphoid tissue.

Slides 6-8: Lymph Node Architecture

Lymph Node - Outer Structure



Capsule - dense connective tissue overlying entire node

Subcapsular sinus - space directly under capsule where **afferent lymphatics/sinuses** drain into

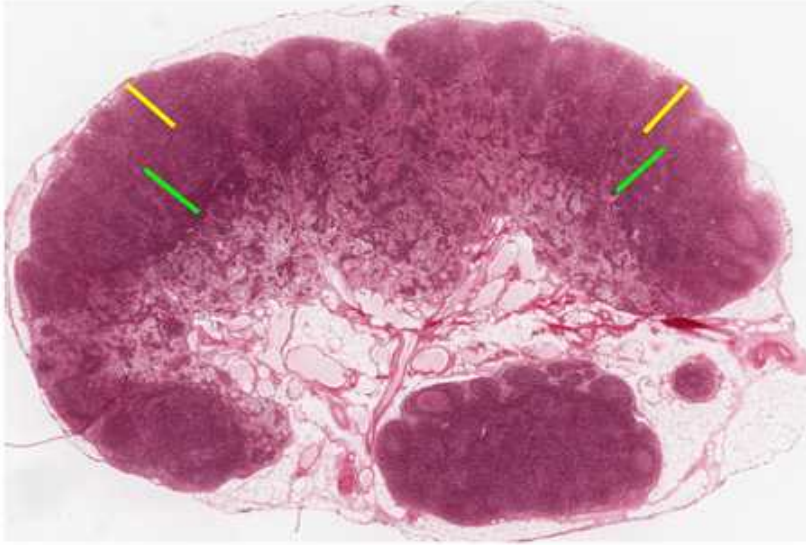
Hilum - Indentation where **efferent lymphatics/sinuses** exit node

Released SAL review slide 6. The slide labels the dense connective-tissue capsule, the subcapsular sinus directly beneath it where afferent lymphatics/sinuses drain, and the hilum where efferent lymphatics/sinuses exit.

OUTER STRUCTURE

Capsule -> subcapsular sinus; hilum = efferent exit.

Lymph Node - Cortex



Outer Cortex

- Many lymphoid nodules of B cells

Deep Cortex (aka Paracortex)

- More diffuse tissue with T cells

Released SAL review slide 7. The slide divides the cortex into an outer cortex with many B-cell lymphoid nodules and a deep cortex/paracortex composed of more diffuse tissue with T cells.

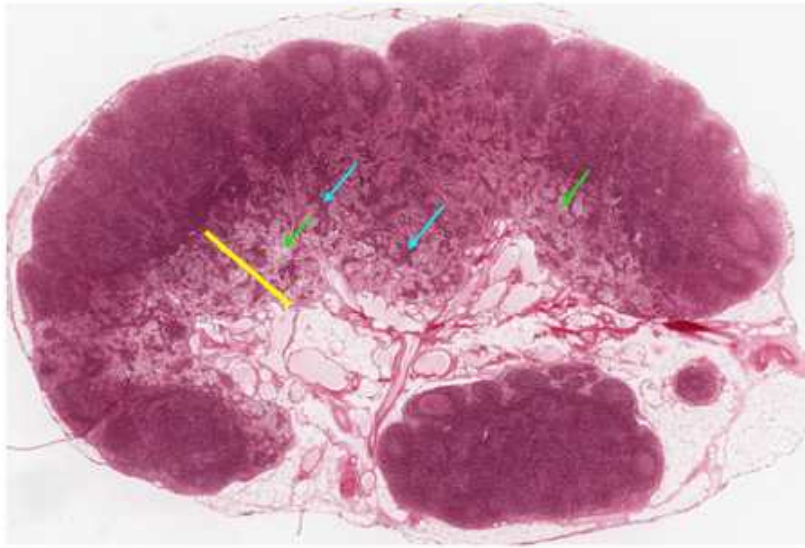
B/T MAP FROM THE REVIEW SLIDE

Outer cortex = B-cell nodules.

Deep cortex / paracortex = diffuse T-cell region.

Ozzy checkpoint - click him once "outer cortex B / paracortex T" is automatic.

Lymph Node - Medulla



Medullary Cords

- Dark staining projections of B cells

Medullary Sinuses

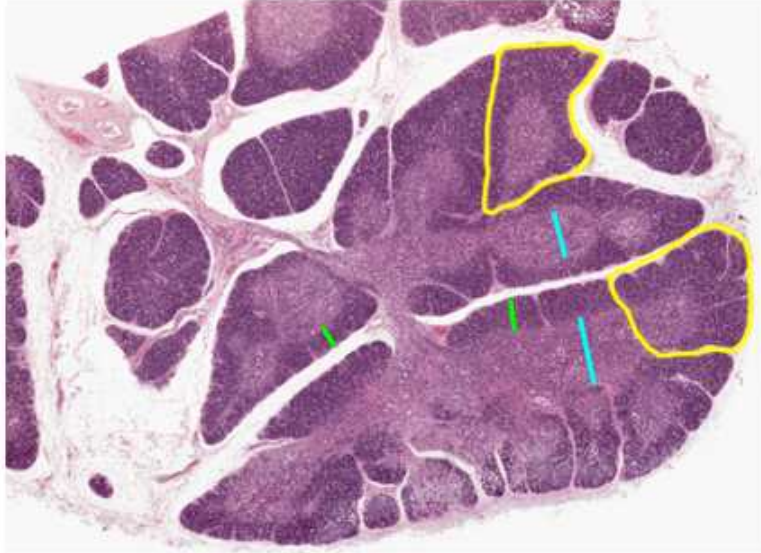
- Light staining channels where lymph flows

Released SAL review slide 8. The slide distinguishes dark-staining medullary cords, labeled as projections of B cells, from light-staining medullary sinuses, labeled as channels where lymph flows.

Node structure	Released-slide cue
Capsule	Dense connective tissue overlying the node
Subcapsular sinus	Space directly under capsule; afferent drainage enters here
Outer cortex	Many lymphoid nodules of B cells
Deep cortex / paracortex	More diffuse tissue with T cells
Medullary cords	Dark-staining projections of B cells
Medullary sinuses	Light-staining channels where lymph flows
Hilum	Indentation where efferent lymphatic/sinus drainage exits

Slides 9-10: Thymus

Thymus



Lymphoid organ with distinct **lobules** and 2 layers

Cortex

- Darker staining, contains immature T cells

Medulla

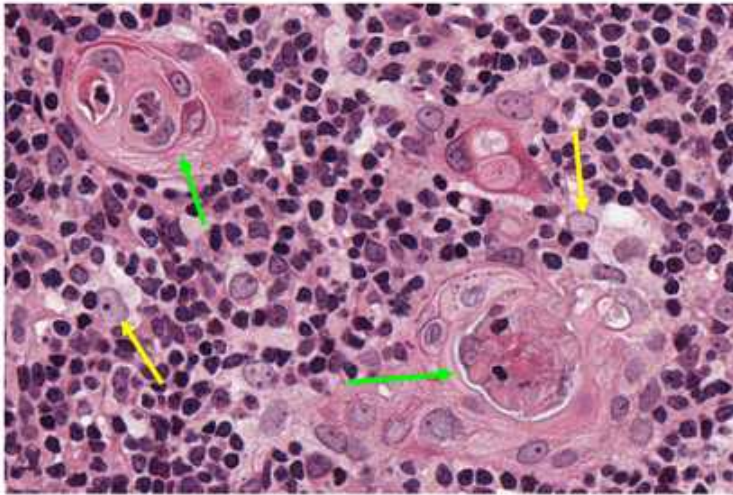
- Lighter staining, contains more mature T cells

Released SAL review slide 9. The review slide identifies the thymus by distinct lobules with two layers: a darker cortex containing immature T cells and a lighter medulla containing more mature T cells.

THYMUS LOW-POWER FINGERPRINT

Lobules + dark cortex (immature T) + light medulla (more mature T).

Thymus - Specialized cells



Epithelial reticular cells

- large flat cells of the stroma around lymphoid cells

Hassall's (Thymic) Corpuscles

- Large keratinized structures in the medulla surrounded by ERCs

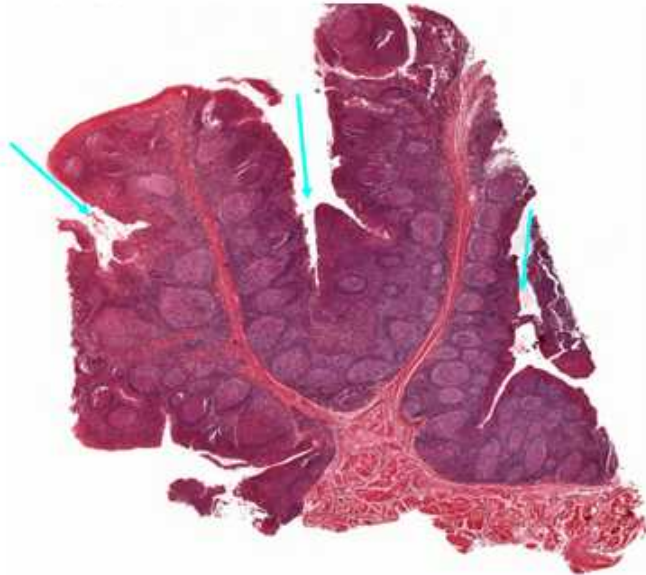
Released SAL review slide 10. The slide labels epithelial reticular cells as large flat stromal cells around lymphoid cells and identifies Hassall's/thymic corpuscles as large keratinized structures in the medulla surrounded by epithelial reticular cells.

HASSALL CORPUSCLE

Large keratinized medullary structure surrounded by epithelial reticular cells.

Slide 11: Tonsil

Tonsil



Many distinct lymphoid nodules

Deep **tonsillar crypts** of
invaginated epithelial tissue

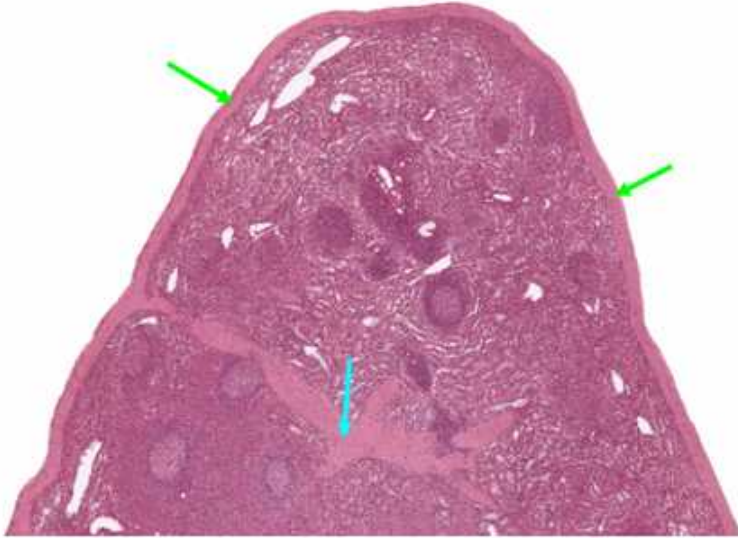
Released SAL review slide 11. The released slide emphasizes many distinct lymphoid nodules and deep tonsillar crypts formed by invaginated epithelial tissue.

TONSIL PRACTICAL FINGERPRINT

Many lymphoid nodules + deep epithelial tonsillar crypts.

Slides 12-14: Spleen

Spleen - Overall Structure



Thick **capsule** with **trabeculae** extending into organ

- Contain branches of splenic artery known as **trabecular arteries**

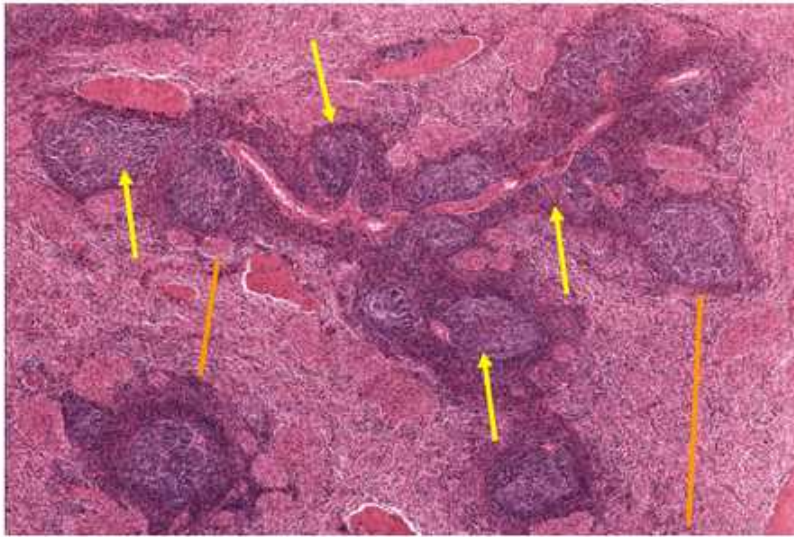


Released SAL review slide 12. The review slide shows a thick capsule with trabeculae extending into the organ; trabeculae contain branches of the splenic artery called trabecular arteries.

SPLEEN OUTER ARCHITECTURE

Thick capsule -> trabeculae -> trabecular arteries.

Spleen - Overall Structure



Red Pulp

- Less dense, lighter staining
- Contains splenic sinuses and surrounding cords

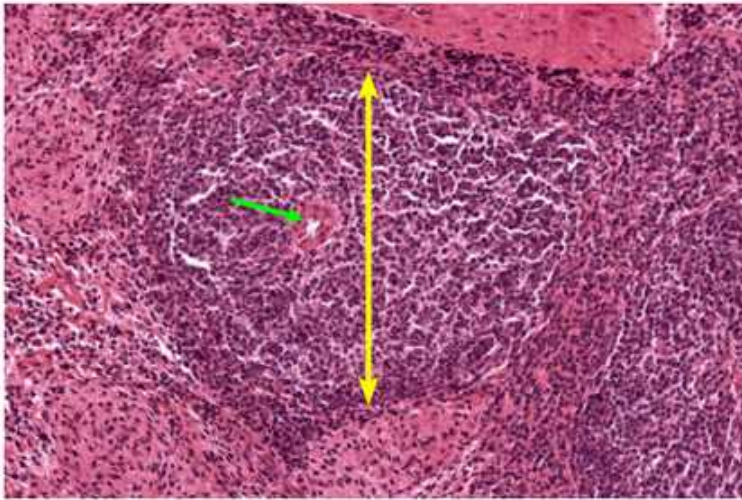
White Pulp

- More dense, darker staining
- Contains diffuse and nodular lymphoid tissue

Released SAL review slide 13. The slide distinguishes less dense, lighter red pulp containing splenic sinuses and surrounding cords from denser, darker white pulp containing diffuse and nodular lymphoid tissue.

Region	Released-slide appearance / contents
Red pulp	Less dense, lighter staining; splenic sinuses + surrounding cords
White pulp	More dense, darker staining; diffuse + nodular lymphoid tissue

Spleen - White Pulp



Central Arteriole

- Branch of trabecular arteriole
- Surrounded by **Periarterial Lymphatic Sheath (PALS)**
 - Diffuse lymphatic tissue

Released SAL review slide 14. The slide identifies the central arteriole as a branch of a trabecular arteriole and shows it surrounded by the periarterial lymphatic sheath, or PALS, described as diffuse lymphatic tissue.

WHITE-PULP LANDMARK

Central arteriole + surrounding PALS.

Official Objective Gap: Pathologic Lymphadenopathy

The released Lymphoid Histology review deck contains normal histology only. It does not show reactive follicular hyperplasia, paracortical hyperplasia, suppurative/granulomatous lymphadenitis, lymphoma, or any other pathologic lymphadenopathy specimen. Because there is no dedicated Word companion either, the earlier pre-SAL reactive-pattern list is **not carried forward as released-slide content.**

WHAT THIS MEANS FOR STUDYING

For this updated SAL guide, know the normal architecture extremely well. If a pathologic lymph-node image appears on an assessment, the most defensible first step from the released materials is to ask whether the normal capsule/cortex/paracortex/medulla pattern is preserved or altered. The released deck does not support a more specific instructor-authored pathology list.

One-Glance Organ Recognition

Structure	Released-slide fingerprint
Diffuse lymphatic tissue	Scattered dark lymphocyte nuclei with scant cytoplasm
Primary nodule	Organized lymphocytes + reticular CT; no pale center shown
Secondary nodule	Pale germinal center
Peyer's patch	Ileal mucosa + nodules + diffuse lymphoid tissue
Lymph node	Capsule/subcapsular sinus; B outer cortex; T paracortex; cords/sinuses; hilum
Thymus	Lobules, dark cortex, light medulla, Hassall corpuscles
Tonsil	Many nodules + deep tonsillar crypts
Spleen	Thick capsule/trabeculae; red vs white pulp; central arteriole + PALS

LAST 60 SECONDS

Primary nodule: no pale GC. **Secondary:** pale GC.

Node: B outer cortex / T paracortex / dark cords / light sinuses.

Thymus: dark immature-T cortex / light more-mature-T medulla / Hassall corpuscles.

Tonsil: deep crypts. **Peyer's:** ileum. **Spleen:** red-white pulp + central arteriole/PALS.

Updated source hierarchy: Schoenlein Lymphoid Histology review slides are primary. No dedicated lymphoid-tissue Word companion exists. Pathologic lymphadenopathy examples are not present in the released review deck.